

Full Stack Development with MERN

Project Documentation format

1. Introduction

- **Project Title:** Online Complaint Registration and Management System
- **Team Members:** Abhilasha bayya

2. Project Overview

- **Purpose:**

The Online Complaint Registration and Management System is a web-based application developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). It enables users to register complaints online, track complaint status, and receive timely updates. Administrators can assign complaints to agents, while agents can resolve assigned complaints efficiently. The system improves transparency, minimizes paperwork, and ensures secure complaint management through role-based authentication.

- **Features:** User Registration & Login
- JWT Authentication
- Role-Based Access Control (User, Admin & Agent)
- Complaint Registration
- Complaint Tracking
- Complaint Status Updates
- Complaint Categories
- Admin Dashboard
- Agent Dashboard
- Complaint Assignment
- Complaint Resolution
- Feedback & Rating System
- User Profile Management
- RESTful API Architecture
- Secure Authentication & Authorization
- Responsive User Interface

3. Architecture

Frontend Architecture (React.js)

The frontend is developed using React.js following a component-based architecture. The application is a Single Page Application (SPA), providing a responsive and interactive interface. React Router DOM is used for navigation, while Context API manages authentication and user session.

Frontend Technologies

- React.js
- React Router DOM
- Bootstrap
- HTML5
- CSS3
- Fetch API

Frontend Modules

- Home Page
- Login Page
- Registration Page
- User Dashboard
- Admin Dashboard
- Agent Dashboard
- Complaint Registration
- Complaint Tracking
- Feedback Page
- Edit Profile

Backend Architecture (Node.js & Express.js)

The backend follows the MVC architecture using Node.js and Express.js. REST APIs handle authentication, complaint management, feedback, and administrative operations.

Backend Technologies

- Node.js
- Express.js
- MongoDB
- Mongoose
- JWT
- bcrypt.js
- dotenv
- cors

Backend Modules

- User Controller
- Complaint Controller
- Agent Controller
- Feedback Controller

- Authentication Middleware
- Role Middleware

Database Architecture (MongoDB)

Users Collection

- Name
- Email
- Password
- Role
- Created Date

Complaints Collection

- Title
- Description
- Category
- Status
- Assigned Agent
- User ID
- Feedback
- Rating
- Admin Remark
- Created Date

Agents Collection

- Name
- Email
- Specialization
- Role

Feedback Collection

- Complaint ID
- Feedback
- Rating

4. Setup Instructions

Prerequisites

- Node.js
- MongoDB Atlas
- VS Code
- Git

- npm

Backend

cd backend

npm install

npm start

Backend URL

http://localhost:5000

Frontend

cd frontend

npm install

npm start

Frontend URL

http://localhost:3000

Environment Variables (.env)

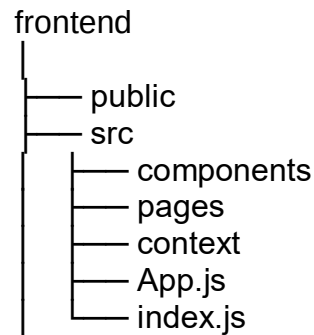
PORT=5000

MONGO_URI=your_mongodb_connection_string

JWT_SECRET=your_secret_key

5. Folder Structure

```
backend
├──
├── config
├── controllers
├── middleware
├── models
├── routes
├── server.js
└── .env
```



6. Running the Application

- **Frontend:** npm start in the client directory.

cd frontend

npm start

- **Backend:** npm start in the server directory

cd backend

npm start

7. API Documentation

User APIs

- POST /register
- POST /login
- GET /profile
- PUT /profile

Complaint APIs

- POST /complaints
- GET /complaints
- PUT /complaints/:id
- DELETE /complaints/:id
- POST /assign-agent

Agent APIs

- GET /agents
- POST /agents

Feedback APIs

- POST /feedback
- GET /feedback

8. Authentication

User Login



JWT Token Generated



Stored in Local Storage



Protected Routes



Authorized API Requests

Security Features

- JWT Authentication
- Password Hashing using bcrypt.js
- Protected Routes
- Role-Based Authorization
- Secure Environment Variables

9. User Interface

User Side

- Registration
- Login
- Dashboard
- Register Complaint
- Track Complaint
- View Feedback
- Edit Profile

Admin Side

- Dashboard
- View Complaints
- Assign Agent
- Edit Complaint
- View Agents

Agent Side

- Dashboard
- Assigned Complaints
- Update Status

UI Features

- Responsive Design
- Bootstrap Cards
- Navbar
- Profile Menu
- Dashboard Statistics
- Status Badges
- Alerts

10. Testing

Testing Strategy

- API Testing using Postman
- Frontend Testing using Browser
- MongoDB Testing
- Authentication Testing
- Authorization Testing
- CRUD Operations Testing

Tested Modules

User Module

- Registration
- Login
- Profile Update

Complaint Module

- Create Complaint
- Edit Complaint

- Delete Complaint
- Status Update

Admin Module

- View Complaints
- Assign Agents
- Manage Complaints

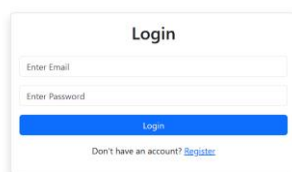
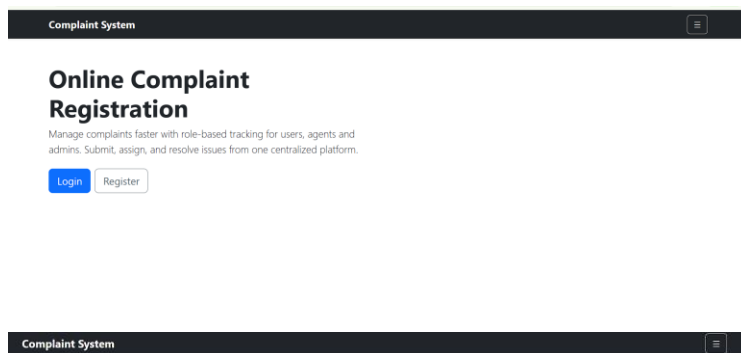
Agent Module

- View Assigned Complaints
- Update Status

Security Testing

- Protected Routes
- Unauthorized Access Prevention
- JWT Verification

11. Screenshots or Demo



Register

Enter Name

Enter Email

Enter Password

Register

Already have an account? [Login](#)

User Dashboard

Create Complaint

Complaint Title

Complaint Description

Complaint Category

Submit Complaint

My Complaints

traffic signal needed

huge amount of accidents as there is no traffic signal

Category: POLICE

Status: **RESOLVED**

Assigned Agent: Kumar

Feedback: No Feedback

Rating: 0

venkat

Home

View Profile

Feedback

Logout

Login

admin@gmail.com

Login

Don't have an account? [Register](#)

Complaint System

Admin Dashboard

Total Complaints10

Total Agents6

Pending0

Resolved5

Agents

Create New Agent

Name

Email

Specialization

Password

Create

Ravi

ravi@gmail.com

ghnc

Edit

Rani

rani@gmail.com

Bechsky

Activate Webhook

Go to Settings to activate Webhook.

water leakage

water pipe broke due to dug up in the middle of the road leading to water wastage

Status: RESOLVED

Assigned Agent: Arushi

Admin Remark: None

Select Agent

Edit Complaint

traffic signal needed

huge amount of accidents as there is no traffic signal

Status: RESOLVED

Assigned Agent: Kumar

Admin Remark: None

Select Agent

Edit Complaint

Assigned Agent: Arushi

Admin Remark: None

Select Agent

Edit Complaint

Select Agent

Ravi

Rani

deepa

Arushi

naresh

Kumar

Select Agent

Edit Complaint

Login

ravi@gmail.com

.....

Login

Don't have an account? [Register](#)

Complaint System

Agent Dashboard

Pending

0

In Progress

3

Resolved

3

Road Damage

Large pothole near college gate

Category: Roads

Status: IN PROGRESS

IN PROGRESS RESOLVED

Road Damage

Large pothole near college gate

Category: Roads

Status: IN PROGRESS

IN PROGRESS RESOLVED

Activate Windows

Go to Settings to activate Windows.

12. Known Issues

- Email notifications are not implemented.
- Complaint attachments are not supported.
- Real-time notifications are unavailable.
- Complaint search filters can be enhanced.

13. Future Enhancements

- Email Notifications
- SMS Alerts
- File Upload with Complaints
- Live Complaint Tracking
- AI-based Complaint Categorization
- Complaint Analytics Dashboard
- Mobile Application (Android/iOS)
- Push Notifications
- Chat Support
- Complaint History Reports
- Admin Analytics Dashboard
- Cloud Deployment